IARS-Based Nanoswimmers Simulation for Gradient Driven Collective Decision Making in Tumor Targeting

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Abstract- This study presents an enhanced computational framework for nanoswimmers navigation based on the Individual Adaptive Regulation Strategy (IARS), integrating gradient sensing, reliability evaluation, and dynamic motion within constrained environments. Simulated biological gradient fields are used to represent tumor-induced signaling landscapes, allowing evaluation under diverse conditions. Each nanoswimmers independently estimates local gradient direction using finite difference approximation and evaluates sensing reliability through a fitness function. High-confidence agents are selected as elites, and their directional information is combined using weighted vector fusion to produce a stable global navigation direction. The framework further incorporates adaptive behavioral modes and constrained movement, enabling realistic trajectory formation and efficient convergence toward target regions. Results demonstrate reduced directional variance, high detection rates, and consistent swarm convergence, confirming the effectiveness of collective intelligence in improving nanoscale navigation. This approach provides a scalable foundation for future autonomous biomedical systems.

Index Terms- Nanoswimmers, Biological Gradient Field, Swarm Intelligence, IARS, Gradient Estimation, Collective Decision Making, Tumor Targeting, Computational Modelling.

I. INTRODUCTION

The Nanoswimmers are tiny engineered agents designed to operate at micro and nanoscale levels within fluid environments, where physical behaviour differs significantly from everyday systems. At this scale, the Reynolds number is extremely low, meaning viscous forces dominate while inertia has minimal influence, making movement slow, constrained, and often affected by random disturbances. These agents must function under challenging conditions such as thermal noise, limited propulsion, restricted sensing range, and the absence of centralized control. However, biological environments naturally contain spatial gradients, including variations in chemical concentration, oxygen levels, and pH. Tumour regions, in particular, create strong and structured biological gradient fields due to abnormal metabolism and irregular vascular activity. These gradients provide meaningful spatial cues, which nanoswimmers can use to guide their movement. In the updated framework, these cues are not only sensed but are actively used to drive navigation, allowing agents to move dynamically through constrained pathways and adapt their trajectories toward target regions in a more realistic manner.

Despite their potential, guiding nanoswimmers effectively remains a complex problem due to several inherent limitations. Each nanoswimmers operates with only local awareness, meaning it has no global understanding of the environment, and its observations are often affected by noise and uncertainty. Relying on a single agent leads to

unstable and inaccurate decisions, especially in dynamic conditions. Additionally, managing large populations becomes difficult when using centralized approaches, as scalability quickly becomes an issue. To address these challenges, the updated system adopts a decentralized and adaptive strategy where multiple agents contribute to decision-making. It introduces reliability-based filtering to prioritize high-confidence agents and incorporates behavioural modes such as exploration and directed pursuit to improve navigation efficiency. Combined with motion constraints that mimic biological pathways, this approach enables more stable, coordinated, and scalable swarm behaviour, significantly improving the ability of nanoswimmers to reach target regions accurately.

II. LITERATURE REVIEW

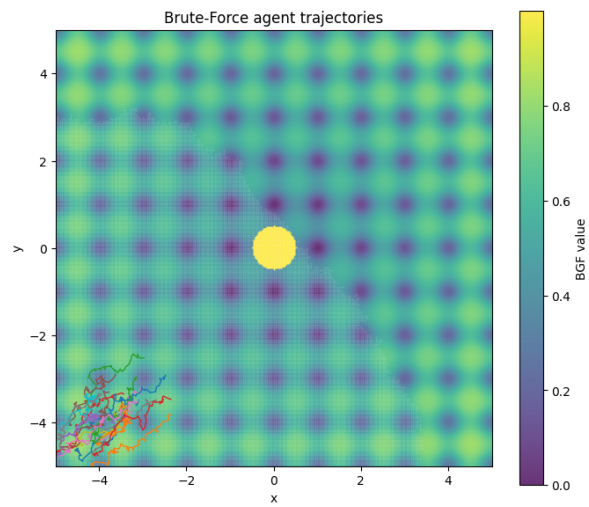
Passive Targeting and Its Practical Limitations

Early nanoswimmers-based targeting approaches mainly relied on passive behaviour, where particles move through the body by following natural physiological processes such as blood flow and diffusion. A commonly cited mechanism is the enhanced permeability and retention effect, where nanoparticles tend to accumulate in tumour regions due to leaky vasculature and poor lymphatic drainage. While this idea appears effective in theory, its practical performance is limited. In real scenarios, only a very small fraction of injected particles actually reach the intended target, largely because their movement lacks direction and control. These particles drift randomly, influenced by surrounding flow and environmental conditions, which makes precise targeting extremely difficult. In complex biological systems, where accurate localization is critical, such randomness significantly reduces effectiveness. As a result, passive targeting, although simple, falls short in reliability and does not meet the demands of advanced therapeutic applications.

Externally Controlled Systems and Their Constraints

To improve upon passive strategies, externally controlled nanoswimmers were introduced, where movement is guided using external forces such as magnetic fields or engineered chemical gradients. These systems provide better directional control, allowing agents to be steered toward specific regions instead of relying entirely on random motion. While this increases targeting accuracy, it introduces several practical limitations. Most approaches require prior knowledge of the target location, which is not always available, especially in early-stage detection. Additionally, the same control signal is typically applied to all nanoswimmers, without considering differences in their positions or the local environment. This uniform control becomes inefficient in heterogeneous biological systems, where conditions vary significantly across regions. Another challenge is the lack of adaptability; these systems often struggle to respond dynamically to changes in the environment. Although externally guided approaches improve directionality, they remain limited in flexibility, scalability, and real-time responsiveness.

Swarm Intelligence Approaches and Remaining Gaps



More recent approaches shift the focus from individual agents to coordinated swarm behaviour, drawing inspiration from natural systems where simple entities collectively achieve complex tasks. In this context, tumour localization can be treated as an optimization problem, where nanoswimmers explore the environment and attempt to converge toward regions of higher signal intensity. The Individual Adaptive Regulation Strategy (IARS) builds on this idea by enabling agents to estimate local gradients, evaluate the reliability of their observations, and contribute to a collective decision based on

confidence. The updated framework extends this further by introducing dynamic motion, adaptive behavioural modes such as exploration and directed pursuit, and movement constraints that reflect vascular pathways. These additions allow nanoswimmers not only to estimate directions but also to physically navigate toward targets in a more realistic manner. Despite these advancements, challenges such as noisy sensing, uneven agent distribution, and environmental complexity still affect performance. To address this, the system emphasizes elite-driven collective estimation, where only the most reliable agents influence the final decision, resulting in more stable, accurate, and scalable navigation across diverse biological conditions.

III. METHODOLOGY

System Architecture and Information Flow

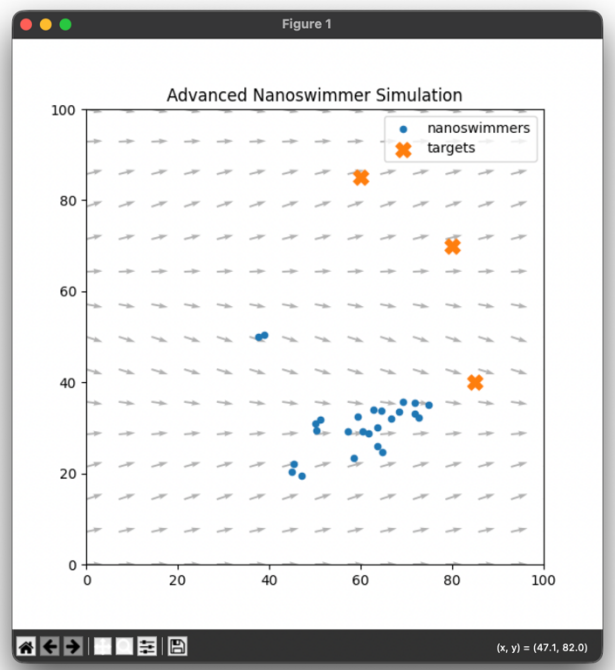
The system is designed as a structured computational pipeline where information flows through multiple stages and is progressively refined at each step. The overall process follows the sequence: Environment → Agents → Gradient Estimation → Fitness Evaluation → Elite Selection → Fusion → Motion. The environment represents the spatial field containing the signal, while agents correspond to nanoswimmers distributed within this space. During gradient estimation, each agent attempts to understand how the signal varies locally around its position. Fitness evaluation assigns a reliability score based on signal strength, indicating how trustworthy each agent's observation is. Elite selection filters out weaker agents and retains only those with high confidence, and fusion combines their directional inputs to generate a stable global estimate. In the updated system, this pipeline is further extended by incorporating motion and behavioural adaptation, allowing agents not only to estimate direction but also to actively move within constrained pathways.

Biological Gradient Field Modelling

$U(x,y)=1-\alpha(x^2+y^2)$

In this formulation, $U(x,y)$represents the signal intensity at a given spatial position, while (x, y)defines the coordinates within the environment. The parameter αcontrols how quickly the signal decreases as the distance from the centre increases. The origin $(0, 0)$corresponds to the tumour location, where the signal is strongest. As the squared distance $x^2 + y^2$increases, the signal gradually weakens, creating a smooth radial gradient. In the updated framework, this basic model is complemented by additional gradient functions such as Matyas, Ackley, and Easom, which introduce more complex and realistic signal landscapes. These variations help evaluate how the system performs under different environmental conditions, including multi-modal and sharply peaked gradients. This ensures that the model is not limited to ideal cases but can handle diverse and challenging scenarios.

Finite Difference Gradient Sensing



$\partial U/\partial x \approx [U(x+\delta,y)-U(x-\delta,y)] / 2\delta$

$\partial U/\partial y \approx [U(x,y+\delta)-U(x,y-\delta)] / 2\delta$

$\nabla U(x,y) = (gx, gy)$

$\theta i = \tan^{-1}(gy/gx)$

The objective here is to estimate the local gradient, which indicates the direction of maximum signal increase. Instead of computing exact derivatives, which require full knowledge of the environment, a finite difference approach is used. Each agent samples nearby points using a small step size $\delta$and compares their values to approximate directional changes. The resulting gradient vector (g_x, g_y)captures both the direction and magnitude of signal increase, while the

angle $\theta_i$provides a directional representation. This method relies only on local information, making it efficient and scalable. It also mirrors biological chemotaxis, where cells respond to concentration differences. In the updated system, this sensing step directly influences motion, meaning that estimated gradients are no longer just theoretical outputs but actively guide nanoswimmer trajectories, making the process more realistic.

Fitness Evaluation and Reliability Meaning

$f_i = U(x_i, y_i)$

Fitness represents a numerical measure of how reliable an agent's observation is, based on the signal strength at its location. Higher fitness values indicate that the agent is positioned closer to the target region and is therefore sensing a stronger and clearer signal. These agents are more likely to produce accurate gradient estimates. In contrast, agents in weaker signal regions are more prone to noise and uncertainty. In the updated framework, fitness not only determines reliability but also influences movement behaviour and decision contribution. Agents with higher fitness have a stronger impact on the collective direction and are more likely to guide the swarm, while lower fitness agents adapt by exploring or following emerging paths.

Elite Selection Mechanism

$E = \rho N, 0 < \rho < 1$

$E = \{i \mid f_i \text{ is among top } E\}$

Here, Nrepresents the total number of agents, and $E$is the subset of elite agents selected based on fitness. The parameter ρcontrols the proportion of agents retained. Only those with the highest fitness values are included in this elite set. This selection process ensures that unreliable or noisy estimates are excluded from influencing the final decision. In the updated system, elite agents play an even more critical role, as they not only guide direction but also influence swarm behavior during movement phases. Their dominance helps stabilize trajectories and improves convergence efficiency in complex environments.

Vector Representation and Weighted Fusion

$v_i = (\cos \theta_i, \sin \theta_i)$

$V = \Sigma(i \in E)\, f_i\, v_i$

$\hat{V} = V / \|V\|$

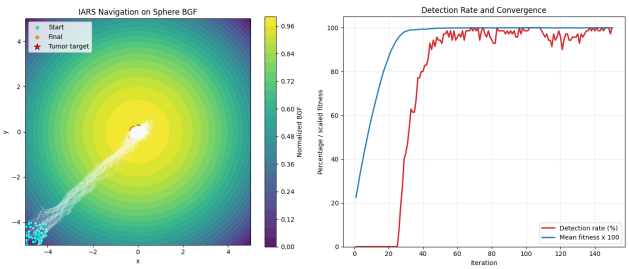
$\theta_{global} = \tan^{-1}(V_y / V_x)$

Each elite agent's directional estimate is converted into a unit vector, allowing directions to be combined geometrically. These vectors are then aggregated using a weighted sum, where fitness values determine the influence of each agent. This ensures that stronger and more reliable signals contribute more significantly to the final direction. The resulting vector is normalized to maintain stability, and the final global direction is obtained from its orientation. In the updated framework, this fused direction is directly used to drive nanoswimmers motion, linking decision-making with physical movement and enabling consistent trajectory formation.

Swarm Intelligence and Simulation Setup

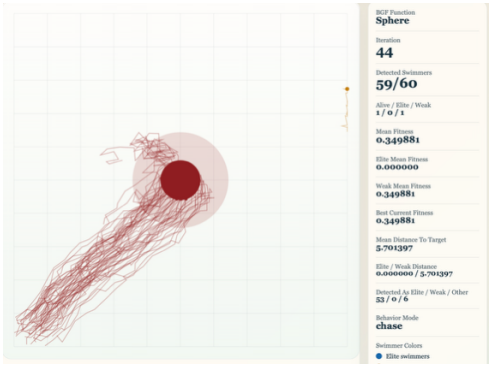
The system is built on swarm intelligence principles, where simple agents operate independently but collectively produce meaningful behaviour. There is no centralized controller; instead, coordination emerges through selective information integration and adaptive responses. This approach reduces variability and improves reliability compared to single-agent systems. The simulation is conducted in a two-dimensional space defined as $[-10,10]^2$, with agents initialized randomly using a uniform distribution. In the updated version, the system extends beyond static sensing to include dynamic motion, behavioral modes such as exploration and directed convergence, and movement constraints that resemble vascular pathways. These additions allow nanoswimmers to actively navigate the environment rather than only estimate direction, resulting in a more complete and realistic simulation of swarm-based tumour targeting.

IV. RESULTS AND FINDINGS



Observed Behaviour and Practical Understanding
When the system is visualized in its updated form, the behaviour becomes much more intuitive and realistic compared to earlier static models. The gradient heatmap still confirms that the underlying field is correctly defined, with signal intensity decreasing smoothly as we move away from the tumour region. However, the key difference now is that nanoswimmers are no longer just estimating directions—they actively move through the environment. The trajectories show how agents gradually align and form clear paths toward the target region, even though their initial movements appear scattered due to local uncertainty and noise. Each nanoswimmers still relies only on nearby information, which introduces variation in early stages, but over time, more reliable agents begin to dominate the overall movement pattern. These higher fitness agents tend to remain closer to the tumour region, where the signal is stronger and clearer. As a result, the swarm naturally organizes itself, and the paths begin to converge. The presence of movement constraints and adaptive behavioural modes further refines this process, ensuring that agents follow more realistic routes rather than moving freely in all directions.

Analytical Interpretation and Accuracy Improvement



Let variance of directions be σ^2-
Variance is a statistical measure that tells us how spread out a set of values is. In this context, it measures how much the direction estimates of different agents differ from each other. A high variance means the directions are scattered and inconsistent, while a low variance means they are closely aligned.

Before selection: high σ^2-
At this stage, all agents are considered, including those with weak or noisy signals. Since many of these agents produce unreliable estimates, the overall spread of directions is large, leading to high variance.

After elite filtering: reduced variance-
Once elite selection is applied, only the agents with high fitness values are retained. These agents are closer to the tumour and experience stronger signals, so their estimates are more consistent. By removing weaker agents, the spread of directions decreases, which means the variance becomes lower.

After fusion: near optimal direction-
After combining the selected agents using weighted fusion, the final direction becomes very close to the true direction toward the tumour. Fusion works by giving more importance to reliable agents, which helps cancel out remaining noise and reinforces correct estimates.
Collective Estimation Accuracy > Individual Estimation Accuracy

This result highlights an important idea. Individual estimation refers to the direction predicted by a single agent, which is often unreliable due to noise and limited information. Collective estimation, on the other hand, combines inputs from multiple reliable agents, leading to a more accurate and stable result. In simple terms, even though each agent makes imperfect decisions, bringing together the best of them significantly improves overall accuracy and reduces error.

V. STRENGTHS & LIMITATIONS

Strengths of the Proposed Approach

One of the key strengths of the updated system is its ability to remain stable and reliable even in the presence of noise, which is unavoidable at micro and nanoscale levels. Instead of depending on a single agent's observation, the framework combines information from multiple nanoswimmers and prioritizes only those with higher reliability, which significantly reduces the effect of random fluctuations. Another major advantage is that the system now goes beyond static estimation and includes actual motion, allowing agents to physically move toward target regions. This makes the model more realistic and better aligned with real-world scenarios. The use of adaptive behavioural modes, such as exploration and directed pursuit, further improves performance by allowing the swarm to adjust its strategy based on the situation. Additionally, the inclusion of movement constraints that mimic vascular pathways adds a layer of biological realism, ensuring that trajectories are not only accurate but also plausible. From a computational perspective, the approach remains efficient, as it relies on simple mathematical operations that can scale well with larger agent populations. The modular pipeline structure also makes it easy to modify or extend individual components, which is valuable for future research and development.

Limitations and Areas for Improvement

Despite these improvements, the system still has several limitations that can be addressed in future work. Although motion is now included, the movement model remains simplified and does not fully capture complex biological factors such as fluid dynamics, varying flow conditions, or interactions with real tissue structures. The gradient fields used in the simulation, while more diverse than before, are still mathematically defined and may not fully represent the irregular and noisy nature of real biological environments. Another limitation is that behavioural switching is based on predefined rules rather than learned or adaptive policies, which may restrict the system's ability to handle highly dynamic scenarios. Additionally, there is still no direct communication between agents, meaning coordination happens indirectly through selection and fusion rather than explicit information sharing. While this keeps the model simple and scalable, incorporating controlled communication could further enhance coordination and efficiency. Addressing these limitations would bring the system closer to real-world applicability and improve its robustness in more complex and unpredictable environments.

VI. REAL-WORLD APPLICATIONS AND SCALABILITY CONSIDERATIONS

The proposed framework demonstrates strong relevance in real-world biomedical applications, particularly in areas such as targeted drug delivery, tumour localization, and precision medicine. In targeted drug delivery, the objective is to ensure that therapeutic agents reach only the affected region while minimizing impact on healthy tissues. The updated system improves this process by enabling nanoswimmers not only to identify the direction of increasing biological signals but also to actively move toward these regions through coordinated trajectories. This allows drug-carrying agents to follow more accurate and stable paths. In tumour localization, the framework can detect and estimate the position of abnormal growths without requiring prior information, as agents dynamically respond to gradient patterns generated by tumour activity. For precision medicine, the inclusion of adaptive behaviour and real-time decision-making makes the system more responsive to local conditions, allowing for more refined and situation-specific intervention strategies.

This approach also aligns closely with the concept of molecular communication, where information exchange occurs through chemical signals rather than traditional electronic means. In this framework, each nanoswimmers operates as an independent sensing and acting unit, capable of interpreting its surroundings and adjusting its behaviour accordingly. The updated model strengthens this concept by incorporating adaptive behavioural modes, where agents can switch between exploration and directed pursuit depending on environmental conditions and sensing reliability. Additionally, the inclusion of movement constraints ensures that nanoswimmers follow realistic pathways similar to biological vessels, making the system more compatible with in vivo conditions. Since there is no reliance on centralized

control, the system remains robust and adaptable even in complex and unpredictable environments, closely reflecting how natural biological systems function.

From a computational perspective, the system remains efficient and scalable despite the added complexity of motion and behaviour. The core processes such as gradient estimation and fitness evaluation still scale linearly with the number of agents, denoted as $O(N)$, since each nanoswimmer performs its own local computations independently. The elite selection and fusion stages operate on a smaller subset of agents, scaling as $O(E)$, where $E$is significantly less than N. Although motion updates and behavioral adjustments introduce additional computation, these operations are lightweight and can be efficiently handled using vectorized processing. This allows multiple agents to be updated simultaneously, reducing computational overhead. As a result, the system can handle large nanoswimmer populations while maintaining performance, making it suitable for large-scale simulations and potential real-world deployment scenarios.

VII. PROBLEM FORMULATION

The problem can be viewed as determining the most reliable direction that guides nanoswimmers toward regions of highest signal intensity within a scalar field $U(x, y)$. This field represents a biological signal landscape where each point in space is associated with a value, and higher values indicate closer proximity to a tumour region. The peak of this field corresponds to the target location. The system consists of Nnanoswimmers, each of which can only sense its immediate surroundings, meaning that no agent has complete knowledge of the entire environment. This local limitation makes the task inherently uncertain and requires intelligent coordination. The objective is to estimate a direction θthat closely matches the true gradient direction θ^*, which points toward the maximum of the field. This can be expressed as minimizing the expected squared error $E[(\theta - \theta^*)^2]$, where the expectation represents the average deviation across agents or observations. In the updated framework, this objective is not only theoretical but is directly tied to movement, as the estimated direction is continuously used to update agent trajectories and guide them toward the target over time.

Rather than depending on a single agent, the system achieves this objective through collective aggregation, where multiple nanoswimmers contribute their local estimates. Each agent provides partial and potentially noisy information, but the system selectively filters and combines only the most reliable inputs. This reduces the impact of uncertainty and leads to a more consistent overall estimate. In the updated model, this collective decision is directly reflected in the movement of agents, where the fused direction influences how the swarm evolves spatially. As a result, the system not only produces a stable directional estimate but also demonstrates actual convergence toward the target region. This highlights an important advantage: while individual agents may make imperfect observations, their coordinated behaviour results in a much more accurate and effective navigation process.

VIII. ALGORITHMIC FRAMEWORK

```
Initialize N nanoswimmers randomly
For each nanoswimmer:
    Sample local environment using finite differences
    Estimate gradient direction θ_i
    Compute fitness f_i

Select top E agents based on fitness

For each elite agent:
    Convert θ_i to vector v_i
    Weight by fitness f_i

Compute global vector:
    V = sum(f_i * v_i)

Return global direction θ_global
```

For each elite agent, the estimated direction $\theta_i$is first converted into a vector v_i, since working with vectors makes it easier to combine directional information in a consistent way. A vector represents direction through its horizontal and vertical components, which avoids the complications that arise when directly averaging angles. Each of these vectors is then scaled using its corresponding fitness value f_i, ensuring that agents with more reliable observations contribute more strongly to the final outcome. The overall direction is obtained by summing these weighted vectors using the expression $V = \sum(f_i \cdot v_i)$, which effectively aggregates the most trustworthy directional cues

across the swarm. This combined vector is then normalized and converted back into a global angle θ_{global}, representing the final navigation direction. In the updated system, this direction is not just computed for analysis but is actively used to guide nanoswimmer movement, resulting in smoother trajectories and more consistent convergence toward the target region, even in the presence of noise and local uncertainty.

IX. ANALYSIS, EVALUATION & PRACTICAL CONSIDERATIONS

Computational Complexity

The efficiency of the system can be understood by examining how computation scales as the number of nanoswimmers increases. Gradient estimation remains an $O(N)$operation, as each agent independently evaluates its local environment. Similarly, fitness evaluation also scales linearly, since every agent computes its own reliability score. The elite selection step involves ranking agents based on fitness, which typically requires $O(N\log N)$time due to sorting. The fusion stage operates only on the selected subset of elite agents, resulting in a complexity of $O(E)$, where $E$is significantly smaller than N. In the updated framework, additional computations such as motion updates and behavioral adjustments are introduced, but these are lightweight and scale linearly as well. The use of vectorized operations ensures that multiple agents can be processed simultaneously, improving execution speed and making the system suitable for large-scale simulations.

Comparison with Existing Approaches

When compared to simpler methods, the advantages of the updated approach become more evident. Single-agent estimation is highly unreliable because it depends entirely on one source of information, which is often noisy. Unfiltered swarm averaging improves data usage but tends to amplify noise by treating all agents equally. In contrast, the proposed system filters out unreliable agents and prioritizes high-quality information through fitness-based selection. Additionally, the inclusion of motion and adaptive behaviour allows the system to not only estimate direction but also physically navigate toward targets. This combination of selective filtering and dynamic movement leads to more stable trajectories and improved convergence compared to traditional approaches.

Error Sources in the System

Several sources of error still affect the system, even in its updated form. Gradient approximation errors occur because agents rely on local sampling rather than exact derivatives. Environmental noise introduces fluctuations in signal measurements, which can distort direction estimates. Agent positioning also plays a role, as those located far from the target receive weaker signals and produce less reliable outputs. In the extended model, movement constraints and path limitations can introduce additional variability, as agents must navigate through restricted regions rather than moving freely. These factors collectively contribute to uncertainty in both estimation and movement.

Error Mitigation Strategies

To handle these challenges, the system incorporates multiple layers of error reduction. Fitness-based filtering ensures that only agents with strong and reliable signals are selected, effectively reducing the influence of noisy inputs. Weighted fusion further enhances accuracy by giving more importance to high-confidence agents. In the updated framework, behavioural adaptation also plays a role in mitigating error, as agents can adjust between exploration and directed movement based on local conditions. Together, these mechanisms improve both the stability of direction estimation and the consistency of physical trajectories, resulting in more reliable overall performance.

Ablation Study Insights

The importance of each component can be understood through ablation analysis. Removing elite selection leads to increased noise, as unreliable agents are no longer filtered out. Eliminating weighting causes instability, since all agents contribute equally regardless of reliability. Without fusion, there is no effective way to combine information, reducing the system to isolated decision-making. In the updated model, removing motion would prevent actual convergence, while disabling behavioural switching would limit adaptability in dynamic environments. These observations confirm that each component, including the newly added motion and behaviour

mechanisms, plays a critical role in overall system performance.

Visualization and Interpretation

Visualization remains a key tool for understanding system behaviour. Heatmaps are used to represent gradient fields, helping verify that signal distributions are correctly defined. Directional vectors illustrate how individual agents interpret the environment, highlighting variability due to local noise. In the updated system, trajectory plots provide deeper insight by showing how agents move over time and gradually converge toward the target. The final fused direction is reflected in the overall movement pattern of the swarm, making it easier to observe how collective intelligence translates into coordinated navigation. These visualizations offer strong qualitative validation of both estimation and motion.

Real World Deployment Challenges

Although the system shows strong performance in simulation, several challenges remain for real-world deployment. Biological fluid dynamics, which influence how particles move in the body, are not fully modelled and could significantly affect trajectories. Energy limitations of nanoswimmers may restrict sensing and computation capabilities. Real biological environments are also more complex and noisy than simulated fields, introducing additional uncertainty. The immune system may interact with nanoswimmers, potentially disrupting their behaviour. While the updated model improves realism by including motion and constraints, further work is needed to bridge the gap between simulation and practical application.

Ethics and Reproducibility

From an ethical perspective, careful consideration is required to ensure safe and responsible use of such systems. Applications within the human body must avoid unintended side effects, such as damage to healthy tissue or interference with natural biological processes. There is also a need to prevent misuse outside medical contexts. In terms of reproducibility, the system is designed to be easily implemented using standard computational tools. The simulation relies on Python, with NumPy for numerical operations and Matplotlib for visualization. Random initialization ensures diversity in agent behaviour, while the deterministic structure of the pipeline allows consistent results under the same conditions. This makes the framework accessible for further research and validation.

X. CONCLUSION

This work establishes and validates a comprehensive nanoswimmers framework based on an enhanced Individual Adaptive Regulation Strategy (IARS), demonstrating how simple decentralized components can collectively produce reliable and goal-directed behavior. By integrating gradient sensing, fitness evaluation, elite selection, weighted vector fusion, and dynamic motion, the system effectively transforms noisy and limited local observations into both a stable global navigation direction and consistent physical convergence toward target regions. Each stage contributes meaningfully, where sensing captures local environmental variations, fitness identifies the most reliable agents, selection filters out weak or uncertain inputs, and fusion combines high-quality information into a coherent direction. The addition of movement, adaptive behavioral modes, and environmental constraints further ensures that these decisions are not only accurate in theory but also translate into realistic navigation paths. The results demonstrate that collective estimation, when combined with motion and selective filtering, significantly improves accuracy, stability, and convergence compared to individual decision-making, highlighting the strength of decentralized and biologically inspired approaches for nanoscale navigation in complex environments.

XI. FUTURE WORK

Future extensions of this work can move beyond decision making and focus on full navigation by incorporating motion execution and control, where nanoswimmers actively move based on the computed direction and continuously update their path. This would involve adding trajectory planning and control mechanisms so that agents can adjust their movement in response to changing signals rather than just estimating direction at a fixed point. Another important direction is the use of clustering techniques such as DBSCAN, which is a density based algorithm that groups nearby agents based on how closely they

are positioned. This can help identify regions where many reliable agents are concentrated, allowing the system to prioritize those areas and improve collective decision making.

The framework can also be enhanced by introducing adaptive behaviours, where agents adjust their internal parameters dynamically instead of following fixed rules. For example, the number of elite agents, the weighting strategy, or the sensing range could change based on environmental conditions, making the system more flexible and responsive. This is important in real scenarios where conditions are rarely constant. In addition, incorporating dynamic biological environments would significantly improve realism. Instead of assuming a static gradient field, the model can be extended to handle time varying conditions, where the signal changes due to factors like tumour growth, fluid flow, or biochemical interactions. This would allow the system to continuously adapt and make decisions in evolving environments, bringing it closer to practical real world applications.

DISCLAIMER

This work is intended solely for educational and research purposes and is designed to explore computational modelling and theoretical concepts related to nanoswimmers systems. It does not represent a clinically validated solution or a medically approved technology. Any discussion of biomedical applications is purely conceptual and should not be interpreted as medical advice, diagnosis, or treatment guidance.

APPENDIX

Simulation Parameters

The system uses N nanoswimmers placed in a 2D continuous space $[-10,10] \times [-10,10]$. The parameter α controls how quickly the signal decays, δ defines the local sampling step for gradient estimation, and ρ determines the fraction of elite agents selected. These parameters shape how the system senses, filters, and makes decisions.

Gradient Function Definition

$U(x,y)=1-\alpha(x^2+y^2)$

This function models the signal field, with maximum intensity at the centre and smooth decay outward. The term x^2+y^2 represents distance from the centre, ensuring a stable and continuous gradient for reliable estimation.

Vector Fusion Representation

$v_i = (\cos \theta_i, \sin \theta_i)$

$V = \Sigma(i \in E)\ f_i\ v_i$

Each elite agent converts its direction into a vector, and all such vectors are combined using fitness based weighting. This produces a single global direction that reflects the most reliable collective estimate.

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